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Integral Concurrent Learning-Based Accelerated Gradient Adaptive Control of Uncertain Euler-Lagrange Systems

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Abstract*—Recent results in the adaptive control literature have made connections to methods in optimization and have led to new adaptive update laws based on accelerated gradient methods. Accelerated gradient methods such as Nesterov's accelerated gradient in numerical optimization have been shown to yield faster convergence than standard gradient methods. However, these results either assume available measurements of the regression error or do not guarantee convergence of the parameter estimation error unless the restrictive persistence of excitation condition is satisfied. In this paper, a new integral concurrent learning (ICL)-based accelerated gradient adaptive update law is developed to achieve trajectory tracking and real-time parameter identification for general uncertain Euler-Lagrange systems. The accelerated gradient adaptation is a higher-order scheme composed of two coupled adaptation laws. A Lyapunov-based method is used to guarantee the closed-loop error system yields global exponential stability under a less restrictive finite excitation condition. A comparative simulation study is performed on a two-link robot manipulator to demonstrate the efficacy of the developed method. Results show the higher-order scheme outperforms standard and ICL-based adaption by 19.6% and 11.1%, respectively, in terms of the root mean squared parameter estimation errors.

I. Introduction

Adaptive methods are a common technique used in nonlinear systems to achieve a control objective, such as trajectory tracking, while also learning parametric model uncertainties in real-time [1]–[3]. Classical adaptive methods typically achieve the control objective, but the parameter estimates may not converge to the true parameters. To guarantee parameter estimates converge to the true parameters, persistence of excitation (PE) is often required [2]–[4]. The PE condition is a restrictive assumption that requires sufficient excitation of the system for all time, and the condition is generally not verifiable online for a nonlinear system.

To eliminate the restrictive PE condition and guarantee convergence of the parameter estimation errors, less restrictive initial excitation conditions have been developed in [5]–[8]. Similarly, results in [9] and [10] exploit a finite excitation (FE) condition that requires the system to be initially excited for a finite amount of time. Moreover, the FE condition can be verified online by examining the minimum singular value of a function of the regressor matrix. Results in [9] and [10] developed a data-driven method called concurrent learning (CL) that is adopted in this paper. Concurrent to real-time execution, sampled input-output data of the system is collected and stored. These CL methods require state derivative measurements that are generally not available and are required to be generated numerically or estimated as in [11]. To eliminate the dependence on unmeasurable state derivatives, results in [12] developed an integral form of the CL method (ICL). Before the FE condition is met, the parameter estimates may not approach the true parameters. Moreover, the parameter estimates may exhibit poor transient performance characterized by oscillatory behavior and slow convergence [13], [14].

Similarly, first-order methods in numerical optimization algorithms are known to yield poor transient performance [15].[1] To improve transient performance in parameter estimates in an optimization setting, a higher-order accelerated gradient method was developed by Nesterov in the seminal works in [16]. Nesterov's method has gained significant interest due to its faster convergence properties compared to first-order methods. To gain insights on algorithm design and analysis, recent developments have bridged connections between discrete-time and continuous-time analogs of Nesterov's method. For example, results in [17] derive continuous-time analogs of the accelerated gradient method. Further results in [18] and [19] use a variational method to derive continuous-time analogs, and provide insights on analysis using Lyapunov methods.

Motivated by the potential for improved convergence properties from accelerated gradient methods, results such as [20] and [21] developed higher-order continuous-time algorithms for real-time parameter estimation in linear regression models. The higher-order schemes are composed of two coupled adaptation laws modeled by two first-order differential equations. Results in [21] use accelerated hybrid dynamics with CL for real-time parameter estimation. Although results in [21] guarantee exponential convergence in the parameter estimation error, this does not apply to general nonlinear control systems.

Higher-order adaptive update laws have been developed for model-reference adaptive control [20] and for general uncertain nonlinear systems [22]. In [20], the linear regression

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[1]First-order optimization methods can be characterized by a recursive difference equation driven by a cost function's gradient. In higher-order accelerated methods, a weighted sum from the previous iteration is used to add a momentum-based term and accelerate convergence.

model assumes the regression error is measurable, which is often unknown in general nonlinear systems. The results in [22] developed a higher-order composite adaptive update law which is driven by the tracking error and a prediction error that is a measurable form of the regression error. Although [20] and [22] guarantee asymptotic convergence of the tracking error, the parameter estimates are not guaranteed to converge to the true parameters. Moreover, it is not clear whether the tracking error convergence is uniform or exponential.

In this paper, a new higher-order ICL-based control and adaptive update law is developed for a general class of uncertain Euler-Lagrange systems. A torque-filtering method is used to reconstruct a measurable form of the regression error. The torque-filtering method eliminates the dependency on unknown state derivatives that are required to compute and store sampled data for ICL. The higher-order adaptation laws are designed as two coupled first-order differential equations. Unlike previous ICL results, the coupling introduces ICL cross-terms in the stability analysis that challenges the ability to obtain parameter convergence. To compensate for the complexities of higher-order adaptation, auxiliary estimates and regression errors are introduced into the control input and adaptation laws. A Lyapunov-based stability analysis guarantees global asymptotic tracking. Furthermore, if a mild FE condition is satisfied, the developed higher-order ICL-based adaptive scheme yields global exponential convergence in the tracking error and parameter estimation errors. To demonstrate the efficacy of the developed method, simulations were conducted on a two-link robot manipulator.

The rest of the paper is organized as follows. Section II introduces an Euler-Lagrange dynamic model and the control objective. Section III introduces the control input and higher-order ICL-based adaptive update laws. A Lyapunov-based stability analysis is used to prove the tracking and parameter identification objectives are achieved in Section IV. Comparative numerical simulations on a two-link robot manipulator are provided in Section V.

II. Problem Formulation

A. Notation

Let $\mathbb{R}$ and $\mathbb{Z}$ denote the set of real numbers and integers, respectively. Let $\mathbb{R}_{\geq 0} \triangleq [0, \infty)$ and $\mathbb{R}_{>0} \triangleq (0, \infty)$ denote the set of positive and strictly positive real numbers, respectively. Similarly, let $\mathbb{Z}_{\geq 0}$ and $\mathbb{Z}_{>0}$ denote the set of positive and strictly positive integers, respectively. For $n, m \in \mathbb{Z}_{>0}$, let $\mathbb{R}^{n\times m}$ and $0_{n\times m}$ denote the space of $n \times m$ dimensional matrices and the zero matrix, respectively. The minimum and maximum eigenvalue of a matrix $A \in \mathbb{R}^{n\times n}$ is denoted by $\lambda_{min}(A) \in \mathbb{R}$ and $\lambda_{max}(A) \in \mathbb{R}$, respectively. The Euclidean norm of a vector $x \in \mathbb{R}^n$ is denoted by $\|x\| \triangleq \sqrt{x^T x}$. The vector space of essentially bounded Lebesgue measurable functions is denoted by $\mathscr{L}_\infty$.

B. Dynamic Model

Consider a general uncertain nonlinear Euler-Lagrange system modeled as [23, Ch. 2]

$$M(q)\ddot{q} + V_m(q,\dot{q})\dot{q} + G(q) + F\dot{q} = \tau, \tag{1}$$

where $q, \dot{q}, \ddot{q} : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ denotes the generalized position, velocity, and acceleration, respectively, $M : \mathbb{R}^n \to \mathbb{R}^{n\times n}$ denotes a generalized inertia matrix, $V_m : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^{n\times n}$ denotes a generalized centripetal-Coriolis matrix, $G : \mathbb{R}^n \to \mathbb{R}^n$ represents a generalized vector of potential forces, $F \in \mathbb{R}^{n\times n}$ represents generalized dissipation effects, and $\tau : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ denotes the control input. The subsequent development is based on the assumption that q and $\dot{q}$ are measurable. The dynamic model in (1) satisfies the following properties and assumption [23, Ch. 2].

Property 1. *The inertia matrix $M(q)$ for all $q \in \mathbb{R}^n$ is positive-definite and satisfies $m_1 \|\xi\|^2 \leq \xi^T M(q)\xi \leq m_2 \|\xi\|^2$ for all $\xi \in \mathbb{R}^n$, where $m_1, m_2 \in \mathbb{R}_{>0}$ denote known constants.*

Property 2. *The inertia and centripetal-Coriolis matrices satisfy the following skew-symmetric relation $\xi^T \left(\dot{M}(q) - 2V_m(q,\dot{q})\right)\xi = 0$ for all $q, \dot{q}, \xi \in \mathbb{R}^n$.*

Assumption 1. *The uncertain dynamics in* (1) *are linear-in-the-parameters and can be expressed as*

$$M(q)\ddot{q} + V_m(q,\dot{q})\dot{q} + G(q) + F\dot{q} = \Psi(q,\dot{q},\ddot{q})\theta^*, \tag{2}$$

for all $q, \dot{q}, \ddot{q} \in \mathbb{R}^n$, where $\Psi : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^{n\times m}$ denotes a known regressor matrix, and $\theta^* \in \mathbb{R}^m$ denotes a vector of unknown constant parameters.

C. Control Objective

The control objective is to track a user-defined desired trajectory $q_d : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ despite parametric uncertainties in the dynamic model. The desired trajectory and its first two time derivatives are assumed to be continuous and bounded, i.e., $q_d, \dot{q}_d, \ddot{q}_d \in \mathscr{L}_\infty$ for all $t \in \mathbb{R}_{\geq 0}$. Note that the desired trajectory need not be persistently exciting and can be set to a constant for the regulation problem.

To quantify the tracking objective, the tracking error $e : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ and auxiliary tracking error $r : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ are defined as

$$e \triangleq q_d - q, \tag{3}$$

$$r \triangleq \dot{e} + \alpha e, \tag{4}$$

respectively, where $\alpha \in \mathbb{R}_{\geq 0}$ denotes a user-defined parameter. Additionally, the control objective is to simultaneously perform real-time parameter estimation to compensate for the parametric model uncertainties in (1). To quantify the parameter identification objective, the parameter estimation error $\tilde{\theta} : \mathbb{R}_{\geq 0} \to \mathbb{R}^m$ is defined as

$$\tilde{\theta} \triangleq \theta^* - \hat{\theta}, \tag{5}$$

where $\hat{\theta} : \mathbb{R}_{\geq 0} \to \mathbb{R}^m$ denotes the parameter estimate.

III. Control Development

A. Controller and Adaptive Update Laws

This section introduces the controller design and higher-order ICL-based adaptive update laws. To facilitate the subsequent stability analysis, a measurable form of the parameter estimation error is constructed using the torque-filtering method in [4]. Let the filtered control input $\tau_f : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ be defined as

$$\tau_f \triangleq f * \tau, \tag{6}$$

where the filter $f : \mathbb{R}_{\geq 0} \to \mathbb{R}$ is defined as $f \triangleq \beta \exp(-\beta t)$, $\beta \in \mathbb{R}_{>0}$ denotes a user-defined parameter, and $*$ denotes the convolution operator. The filtered control input in (6) can be generated by $\dot{\tau}_f + \beta \tau_f = \beta \tau$ with $\tau_f(0) = 0_n$ [4, Ch. 8.7]. Using (1) and (2), the filtered control input in (6) can be expressed as

$$\tau_f = \Psi_f(q, \dot{q})\,\theta^*, \tag{7}$$

where the filtered regressor $\Psi_f : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^{n \times m}$ is defined as $\Psi_f \triangleq f * \Psi$. The regressor matrix $(q, \dot{q}, \ddot{q}) \mapsto \Psi(q, \dot{q}, \ddot{q})$ depends on the unmeasurable acceleration $\ddot{q}$. To obtain a measurable form of the filtered regressor Ψ_f that is independent of $\ddot{q}$, the regressor matrix can be re-written as $\Psi = \dot{h} + g$, where $(q, \dot{q}, \ddot{q}) \mapsto \dot{h}(q, \dot{q}, \ddot{q})$ and $(q, \dot{q}) \mapsto g(q, \dot{q})$ are known functions. Then the filtered regressor can be expressed as

$$\Psi_f = f * \dot{h} + f * g. \tag{8}$$

To eliminate the dependence on $\ddot{q}$, the following property of convolutions is used [24, Eq. 6.6.7]

$$f * \dot{h} = \dot{f} * h + f(0)\,h - f h(0), \tag{9}$$

where the function $(q, \dot{q}) \mapsto h(q, \dot{q})$ is known and measurable. Let $\psi_{f,1} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^{n \times m}$ and $\psi_{f,2} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^{n \times m}$ be defined as $\psi_{f,1} \triangleq \dot{f} * h$ and $\psi_{f,2} \triangleq f * g$, respectively. Using (8) and (9), the measurable form of the filtered regressor is

$$\Psi_f = \psi_{f,1}(t) + \psi_{f,2}(t) + \beta h(t) - \beta \exp(-\beta t)\,h(0), \tag{10}$$

where $\psi_{f,1}$ can be generated by $\dot{\psi}_{f,1} + \beta \psi_{f,1} = -\beta^2 h$ with $\psi_{f,1}(0) = 0_{n \times m}$, and $\psi_{f,2}$ can be generated by $\dot{\psi}_{f,2} + \beta \psi_{f,2} = \beta g$ with $\psi_{f,2}(0) = 0_{n \times m}$.

Based on the subsequent stability analysis, the controller is designed as

$$\tau \triangleq kr + Y\hat{\theta} + e - 2Y\left(\hat{\theta} - \nu\right), \tag{11}$$

where $k \in \mathbb{R}_{>0}$ denotes a user-defined parameter, $\nu : \mathbb{R}_{\geq 0} \to \mathbb{R}^m$ denotes an auxiliary parameter estimate, and Y denotes a known regressor matrix that is defined subsequently in the closed-loop error development.

Based on the subsequent stability analysis, the higher-order ICL-based adaptive update laws $\dot{\nu} : \mathbb{R}_{\geq 0} \to \mathbb{R}^m$ and $\dot{\hat{\theta}} : \mathbb{R}_{\geq 0} \to \mathbb{R}^m$ are designed as

$$\dot{\nu} \triangleq \Gamma\left(Y^T r + k_1 \sum_{i=1}^{N} \Psi_{f,i}^T \left(\tau_{f,i} - \Psi_{f,i}^T \nu\right)\right), \tag{12}$$

$$\dot{\hat{\theta}} \triangleq -\Gamma\left(k_2\left(\hat{\theta} - \nu\right) - k_1 \sum_{i=1}^{N} \Psi_{f,i}^T \left(\tau_{f,i} - \Psi_{f,i}^T \nu\right)\right), \tag{13}$$

where $k_1, k_2 \in \mathbb{R}_{>0}$ are user-defined parameters, and $\Gamma \in \mathbb{R}^{m \times m}$ denotes a user-defined positive definite learning gain matrix. In (12) and (13), the terms $\tau_{f,i} \triangleq \tau_f(t_i)$ and $\Psi_{f,i} \triangleq \Psi_f(t_i)$ for all $t_i \in \mathbb{R}_{\geq 0}$ are sampled data points at $N \in \mathbb{Z}_{\geq 0}$ discrete time instants that are collected concurrent to real-time execution.[2] The update laws in (12) and (13) are the implementable forms. Using (7), the adaptive update laws in (12) and (13) can also be expressed as

$$\dot{\nu} = \Gamma\left(Y^T r + k_1 \sum_{i=1}^{N} \Psi_{f,i}^T \Psi_{f,i}\left(\theta^* - \nu\right)\right), \tag{14}$$

$$\dot{\hat{\theta}} = -\Gamma\left(k_2\left(\hat{\theta} - \nu\right) - k_1 \sum_{i=1}^{N} \Psi_{f,i}^T \Psi_{f,i}\left(\theta^* - \nu\right)\right), \tag{15}$$

respectively, which will be used in the subsequent stability analysis.[3] In (14) and (15), $\sum_{i=1}^{N} \Psi_{f,i}^T \Psi_{f,i}$ is the ICL term that contains the recorded input-output data generated by the dynamics.

B. Closed-Loop Error System

Taking the time derivative of the auxiliary tracking error in (4), pre-multiplying by $M(q)$, and substituting in (1), the open-loop error system can be expressed as

$$M(q)\,\dot{r} = Y(q, \dot{q}, q_d, \dot{q}_d, \ddot{q}_d)\,\theta^* - \tau - V_m(q, \dot{q})\,r, \tag{16}$$

where the known regressor $Y : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^{n \times m}$ and unknown parameters θ^* are defined based on the relation $Y\theta^* = M(q)(\ddot{q}_d + \alpha \dot{e}) + V_m(q, \dot{q})(\dot{q}_d + \alpha e) + G(q) + F\dot{q}$. Substituting the controller in (11) into (16) yields the closed-loop error system

$$M\dot{r} = -kr + Y\tilde{\theta} - V_m r - e + 2Y\left(\hat{\theta} - \nu\right). \tag{17}$$

IV. Stability Analysis

To facilitate the subsequent stability analysis, the following FE condition is assumed.[4]

Assumption 2. *[12] There exists a time $T \in \mathbb{R}_{>0}$ such that $\lambda_{min}\left(\sum_{i=1}^{N} \Psi_{f,i}^T \Psi_{f,i}\right) \geq \gamma$, where $\gamma \in \mathbb{R}_{>0}$ is a user-defined parameter.*

Before the FE condition is satisfied, the developed controller and adaptive update laws are shown to ensure the closed-loop error system remains bounded and achieves the tracking objective. To verify the FE condition is met, the minimum singular value of the sum of discrete regression matrices is checked to verify it is positive. After Assumption 2 is satisfied, the tracking and parameter estimation errors are shown to be contained within an exponentially converging envelope.

To facilitate the subsequent stability analysis, let $z : \mathbb{R}_{\geq 0} \to \mathbb{R}^{2n+2m}$ denote a concatenated state vector defined as

[2] See [12] for discussions on data collection and implementation of ICL.

[3] The update laws in (14) and (15) are used only for the subsequent stability analysis, whereas the update laws in (12) and (13) are used in implementation.

[4] To facilitate excitation in the system, a transient dither signal can be added to the desired trajectory. To satisfy the FE condition and guarantee the parameter estimation errors converge, the system must only be initially excited for a finite period of time. The FE condition is far less restrictive as the addition of a dither signal to the desired trajectory may be required for all time to satisfy the PE condition.

$z \triangleq \left[e^T, r^T, (\theta^* - \nu)^T, \left(\hat{\theta} - \nu\right)^T\right]^T$. Consider the candidate Lyapunov function $V : \mathbb{R}^{2n+2m} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ defined as

$$V(z,t) \triangleq \frac{1}{2}r^T M r + \frac{1}{2}e^T e + \frac{1}{2}(\theta^* - \nu)^T \Gamma^{-1}(\theta^* - \nu) + \frac{1}{2}\left(\hat{\theta} - \nu\right)^T \Gamma^{-1}\left(\hat{\theta} - \nu\right), \tag{18}$$

which satisfies the inequality

$$c_1 \|z\|^2 \leq V(z,t) \leq c_2 \|z\|^2, \tag{19}$$

where the known constants $c_1, c_2 \in \mathbb{R}_{>0}$ are defined as $c_1 \triangleq \frac{1}{2}\min\left\{1, m_1, \lambda_{min}\left\{\Gamma^{-1}\right\}\right\}$ and $c_2 \triangleq \frac{1}{2}\max\left\{1, m_2, \lambda_{max}\left\{\Gamma^{-1}\right\}\right\}$, where m_1 and m_2 were defined in Property 1.

Theorem 1. *Consider a system modeled by the dynamics in* (1) *that satisfies Properties 1 and 2. Let Assumption 1 hold. The controller in* (11) *and higher-order adaptive update laws in* (14) *and* (15) *ensure the closed-loop error system in* (17) *yields global asymptotic tracking in the sense that* $\lim_{t\to\infty}\|e(t)\| = 0$, $\lim_{t\to\infty}\|r(t)\| = 0$, *and* $\lim_{t\to\infty}\left\|\hat{\theta}(t) - \nu(t)\right\| = 0$ *for all initial conditions* $z(0) \in \mathbb{R}^{2n+2m}$.

Proof. Taking the time derivative of (18), using (4) and (5), applying Property 2, and substituting the closed-loop error system in (17) and adaptive update laws in (14) and (15) yields

$$\dot{V} = -kr^T r - \alpha e^T e - k_2\left(\hat{\theta} - \nu\right)^T\left(\hat{\theta} - \nu\right) - k_1(\theta^* - \nu)^T\left(\sum_{i=1}^{N}\Psi_{f,i}^T\Psi_{f,i}\right)(\theta^* - \nu). \tag{20}$$

Since the term $\sum_{i=1}^{N}\Psi_{f,i}^T\Psi_{f,i}$ is positive semi-definite, (20) can be upper bounded as $\dot{V} \leq -kr^T r - \alpha e^T e - k_2\left(\hat{\theta} - \nu\right)^T\left(\hat{\theta} - \nu\right)$. From (18) and the fact that $\dot{V} \leq 0$, $V \in \mathscr{L}_\infty$, which implies $z \in \mathscr{L}_\infty$, and hence, $e, r, \nu, \hat{\theta} \in \mathscr{L}_\infty$. Using (3)–(5) implies $q, \dot{q}, \dot{\hat{\theta}} \in \mathscr{L}_\infty$, respectively. Using (4), the fact that $e, r \in \mathscr{L}_\infty$ implies $\dot{e} \in \mathscr{L}_\infty$. The fact that $q, \dot{q}, q_d, \dot{q}_d, \ddot{q}_d \in \mathscr{L}_\infty$ implies $Y \in \mathscr{L}_\infty$. Using (17), the fact that $e, r, Y, \hat{\theta}, \nu \in \mathscr{L}_\infty$ implies $\dot{r} \in \mathscr{L}_\infty$. Using (11), the fact that $e, r, Y, \hat{\theta}, \nu \in \mathscr{L}_\infty$ implies the controller $\tau \in \mathscr{L}_\infty$ is bounded. Using (12), the fact that $r, Y, \nu \in \mathscr{L}_\infty$ implies the auxiliary parameter estimate update law $\dot{\nu} \in \mathscr{L}_\infty$ is bounded. Using (13), the fact that $\hat{\theta}, \nu \in \mathscr{L}_\infty$ implies the parameter estimate update law $\dot{\hat{\theta}} \in \mathscr{L}_\infty$ is bounded. Since $e, r, \left(\hat{\theta} - \nu\right) \in \mathscr{L}_\infty$ and $\dot{e}, \dot{r}, \left(\dot{\hat{\theta}} - \dot{\nu}\right) \in \mathscr{L}_\infty$, then $e, r, \left(\hat{\theta} - \nu\right)$ are uniformly continuous. Moreover, using (20) implies $e, r, \left(\hat{\theta} - \nu\right) \in \mathscr{L}_2$. Then by invoking Barbalat's Lemma (i.e, Corollary A.7 in [1]), $\lim_{t\to\infty}\|e(t)\| = 0$, $\lim_{t\to\infty}\|r(t)\| = 0$, and $\lim_{t\to\infty}\left\|\hat{\theta}(t) - \nu(t)\right\| = 0$. ☐

In Theorem 1, the developed controller and adaptive update laws are shown to ensure the closed-loop error system is bounded and achieves the tracking objective. By Assumption 2, there exists a finite period of time in which the FE condition is satisfied. Then exponential convergence bounds on the tracking and parameter estimation errors can be established as in the following theorem.

Theorem 2. *Consider a system modeled by the dynamics in* (1) *that satisfies Properties 1 and 2. Let Assumptions 1 and 2 hold. The controller in* (11) *and higher-order adaptive update laws in* (14) *and* (15) *ensures the equilibrium point* $z = 0_{2n+2m}$ *of the closed-loop error system in* (17) *is globally exponentially stable.*

Proof. Following the proof for Theorem 1, we can obtain (20). By Assumption 2, the minimum eigenvalue of $\lambda_{min}\left(\sum_{i=1}^{N}\Psi_{f,i}^T\Psi_{f,i}\right) \geq \gamma$ is strictly positive for all $t \in [T, \infty)$. Then (20) can be upper bounded as $\dot{V} \leq -k\|r\|^2 - \alpha\|e\|^2 - k_1\gamma\|\theta^* - \nu\|^2 - k_2\left\|\hat{\theta} - \nu\right\|^2$ for all $t \in [T, \infty)$. Then using (18) yields

$$\dot{V} \leq -2\lambda V, \qquad \forall t \in [T, \infty), \tag{21}$$

where $\lambda \triangleq \frac{1}{2c_2}\min(k, \alpha, k_1\gamma, k_2)$ denotes a known constant. Invoking the Comparison Lemma in [25, Lemma 3.4] and using (19) yields

$$\|z(t)\| \leq \sqrt{\frac{c_2}{c_1}}\|z(T)\|\exp(\lambda T)\exp(-\lambda t), \quad \forall t \in [T, \infty). \tag{22}$$

From (19) and the fact that $\dot{V} \leq 0$, $V \in \mathscr{L}_\infty$ implies $\|z(t)\| \leq \sqrt{\frac{c_2}{c_1}}\|z(0)\|$ for all $t \in \mathbb{R}_{\geq 0}$. Then upper bounding (22) yields the exponential convergence bound

$$\|z(t)\| \leq \frac{c_2}{c_1}\|z(0)\|\exp(\lambda T)\exp(-\lambda t), \tag{23}$$

for all $t \in \mathbb{R}_{\geq 0}$ and initial conditions $z(0) \in \mathbb{R}^{2n+2m}$. Hence, the equilibrium point $z = 0_{2n+2m}$ is globally exponentially stable. ☐

V. Simulations

A. Comparison Study

To demonstrate the performance of the developed method, a simulation study was conducted on a two-link planar revolute robot with $M : \mathbb{R}^2 \to \mathbb{R}^{2\times 2}$, $V_m : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^{2\times 2}$, and $F \in \mathbb{R}^{2\times 2}$ modeled as [12]

$$M(q) = \begin{bmatrix} p_1 + 2p_3c_2, & p_2 + p_3c_2 \\ p_2 + p_3c_2, & p_2 \end{bmatrix}, \tag{24}$$

$$V_m(q, \dot{q}) = \begin{bmatrix} -p_3s_2\dot{q}_2, & -p_3s_2(\dot{q}_1 + \dot{q}_2) \\ p_3s_2\dot{q}_1, & 0 \end{bmatrix}, \tag{25}$$

$$F = \begin{bmatrix} f_1, & 0 \\ 0, & f_2 \end{bmatrix}. \tag{26}$$

In (24)–(26), the angular joint state $q : \mathbb{R}_{\geq 0} \to \mathbb{R}^2$ is defined as $q \triangleq [q_1, q_2]^T$, c_2 denotes $\cos(q_2)$, and s_2 denotes $\sin(q_2)$. The nominal parameters of the two-link robot model in (24)–(26) are $p_1 = 3.473$, $p_2 = 0.196$, $p_3 = 0.242$, $f_1 = 5.3$, and $f_2 = 1.1$.

Three simulation experiments were conducted. Each simulation was conducted for 30 s with initial conditions $q(0) = [1.22, -0.52]^T$ rad and $\dot{q}(0) = [0, 0]^T$ rad/sec. The desired trajectory $q_d(t) \triangleq [q_{d1}, q_{d2}]^T$ was selected as

$$q_d \triangleq \begin{bmatrix} \cos(0.5t) \\ 2\cos(t) \end{bmatrix}. \tag{27}$$

The first simulation was performed with a standard gradient-based adaptive design given by [24]

$$\begin{aligned} \tau &\triangleq kr + Y\hat{\theta} + e, \\ \dot{\hat{\theta}} &\triangleq \Gamma Y^T r. \end{aligned} \tag{28}$$

In the second simulation, the ICL adaptive design in [12] was used. The control input was the same as in (28), and the implementable form of the ICL adaptive update law was

$$\dot{\hat{\theta}} \triangleq \Gamma\left(Y^T r + k_1 \sum_{i=1}^{N} \Psi_{f,i}^T \left(\tau_i - \Psi_{f,i}\hat{\theta}\right)\right), \tag{29}$$

which yields the analytic form $\dot{\hat{\theta}} \triangleq \Gamma\left(Y^T r + k_1 \sum_{i=1}^{N} \Psi_{f,i}^T \Psi_{f,i}\tilde{\theta}\right)$. The developed higher-order ICL-based adaptive control design in (11)–(13) was used in the third simulation. In the second and third simulation, data was collected and stored during real-time execution until the FE condition was satisfied. The FE condition was verified online by checking if the minimum eigenvalue condition $\lambda_{min}\left(\sum_{i=1}^{N} \Psi_{f,i}^T \Psi_{f,i}\right) \geq \gamma$ was satisfied, where $\gamma = 1$. The ICL-based terms in the adaptation laws were not executed until the FE condition was satisfied. To provide a comprehensive comparison between the performance from each adaptive control scheme, the user-defined design constants were selected equally across each case where applicable. The parameters used in each case are shown in Table I.

Table I
SIMULATION PARAMETERS

Adaptation Law	Γ	k	k_1	k_2	β	γ
Standard Adaptive in [24]	$20I_{5\times5}$	-	-	-	-	-
ICL Adaptive in [12]	$20I_{5\times5}$	1	0.1	-	0.03	1
Developed Method	$20I_{5\times5}$	1	0.1	0.1	0.03	1

Figure 1 illustrates the evolution of the normalized parameter estimation error trajectories for each case. To compare the performance between the methods, the root mean square (RMS) of the parameter estimation error was calculated. The RMS parameter estimation error corresponding to the standard adaptive, ICL adaptive, and developed method were 3.12, 3.00, and 2.75, respectively. The developed method had the lowest RMS parameter estimation error and is 11.77% and 8.38% lower than the standard adaptive and ICL adaptive methods, respectively. The parameter estimates over the duration of the simulation with the standard adaptive method did not converge to the true parameters, as expected since only tracking error convergence is guaranteed. In both simulations for the ICL adaptive and developed method, the parameter estimates converged to the true parameters at approximately 10 s when the

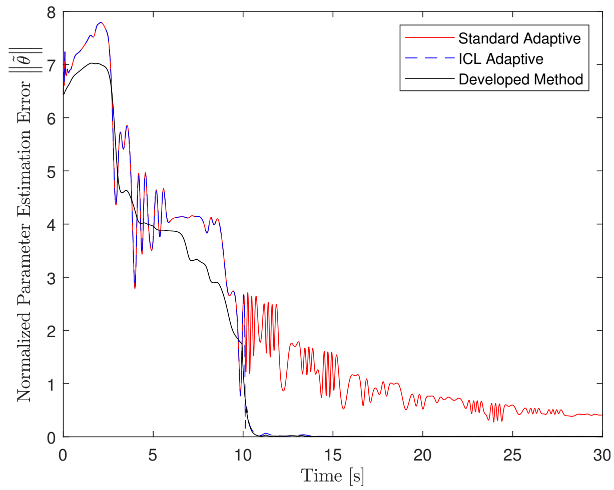

Figure 1. Evolution of the normalized parameter estimation error trajectories $\|\tilde{\theta}\|$ for each simulation. The red line represents the simulation using the standard adaptive method in (28). The blue line represents the simulation using the ICL adaptive method in (28) and (29). The black line represents the simulation using the developed method in (11)–(13).

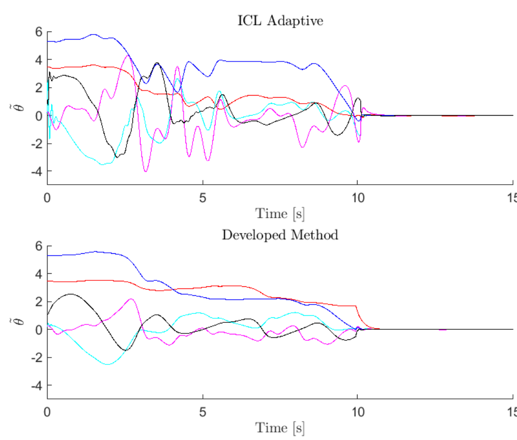

Figure 2. (top): Parameter estimation error $\tilde{\theta}$ using the ICL adaptive method. (bottom): Parameter estimation error using the developed method.

FE condition was satisfied. However, the developed method exhibited improved transient performance compared to the ICL adaptive method, which had high oscillatory behavior. From 0–10 s, the RMS parameter estimation error for the ICL adaptive method was 5.19, whereas the developed method had a 8.29% decreased RMS error of 4.76. Additionally, the ICL adaptive simulation exhibited higher parameter estimation error from approximately 3–6 s than the developed method. To better compare the transient performance, the parameter estimation errors from both the ICL-based method and the developed method are shown in Figure 2.

Figure 3 illustrates the evolution of the normalized tracking error trajectory for each simulation. To compare the tracking performance, the RMS tracking error was computed. The RMS tracking errors corresponding to the simulations for the standard adaptive, ICL adaptive, and developed method are 27.06, 26.99, and 26.53 deg, respectively. The tracking error performance across all simulations were similar. However,

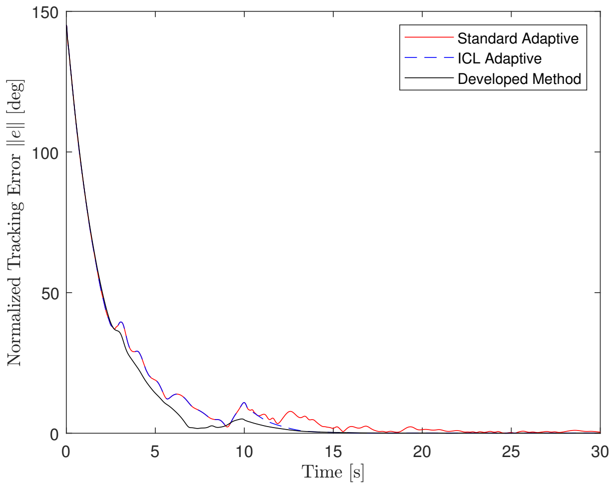

Figure 3. Evolution of the normalized tracking error trajectory $\|e\|$ for each simulation.

the tracking error trajectories in the simulations with the standard adaptive and ICL adaptive methods have higher oscillatory behavior. The control input in (28) is dependent on $\hat{\theta}$. Consequently, oscillatory behavior in the parameter estimates may induce oscillations in the state trajectories, as seen from approximately 3–10 s.

Table II summarizes the simulations results in this section. In the leftmost column, e, e_{0-10}, and e_{ss} denote the RMS of the tracking error, tracking error from 0–10 s, and steady-state tracking error, respectively. Additionally, $\tilde{\theta}$, $\tilde{\theta}_{0-10}$, and $\tilde{\theta}_{ss}$ denote the RMS of the total, transient, and steady-state parameter estimation error, respectively.

Table II
RMS Errors

RMS	Standard Adaptive	ICL Adaptive	Developed Method
e [deg]	27.06	26.99	26.53
e_{0-10} [deg]	46.67	46.67	45.93
e_{ss} [deg]	3.07	1.99	1.03
$\tilde{\theta}$	3.12	3.00	2.75
$\tilde{\theta}_{0-10}$	5.19	5.19	4.76
$\tilde{\theta}_{ss}$	1.05	0.19	0.09

VI. Conclusion

Motivated by the improved transient performance, this paper developed a new higher-order ICL-based adaptive update law. A general uncertain Euler-Lagrange dynamic system was considered. A Lyapunov-based analysis was used to guarantee the tracking and parameter estimation objectives were achieved. Under an FE condition, the closed-loop system yields global exponential stability of the tracking and parameter identification errors. To demonstrate the efficacy of the developed method, comparative simulations were performed on a two-link robot manipulator in Section V. The simulation study showed the developed higher-order ICL adaptation outperformed the standard adaptive and the ICL adaptive schemes by 19.6% and 11.1%, respectively, in terms of the root mean squared parameter estimation errors.

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